Assignment 13.2

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Batch -04

Task 1 – Eliminating Repetitive Logic

PROMPT:
Refactor the following Python code to remove repeated logic.
Create a reusable function to calculate area and perimeter of a rectangle.
Ensure the output remains the same and add proper docstrings.

length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

CODE:

```python
def rectangle_details(length, width):
    """
    Calculates and prints area and perimeter of a rectangle
    """
    area = length * width
    perimeter = 2 * (length + width)

    print("Area:", area)
    print("Perimeter:", perimeter)


rectangle_details(8, 4)
rectangle_details(12, 6)
```

OUPUT:

```
sushruthchada@sushruths-MacBook-Air-2 codir
Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
sushruthchada@sushruths-MacBook-Air-2 codir
```

Task 2 – Modularizing Inline Calculations

PROMPT:

Refactor the Python code by extracting repeated discount calculations into a reusable function with proper documentation.

amount = 1000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)

amount = 2000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)

CODE:

```python
def calculate_final_amount(amount):
    """
    Calculates final amount after 10% discount
    """
    discount = amount * 0.10
    final_amount = amount - discount
    print("Final Amount:", final_amount)


calculate_final_amount(1000)
calculate_final_amount(2000)
```

OUPUT:

```
sushruthchada@sushruths-MacBook-Air-2 coding % /usr/local/bin/python3 "/Users/sushruth
Final Amount: 900.0
Final Amount: 1800.0
sushruthchada@sushruths-MacBook-Air-2 coding %
0 ⚠ 0   ⚡ BLACKBOX Agent   Indexing completed.   Open Website   Generate Commit Message
```

Task 3 – Improving Loop Performance:

PROMPT:

Optimize the following Python code by improving lookup performance.
Use efficient data structures while keeping the output the same.

students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", “Bhavya"]

CODE:

```
competitive programmming > ass4.7.py > ...
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

student_set = set(students)   # faster lookup

for name in check_list:
    if name in student_set:
        print(name, "found")
    else:
        print(name, "not found")
```

OUTPUT:

```
sushruthchada@sushruths-MacBook-Air-2 coding % /usr/local/bin/python3 "/Users/sushruthchada/Downloads/coding/competiti
Charan found
Esha not found
Bhavya found
sushruthchada@sushruths-MacBook-Air-2 coding %
```

Task 4 – Replacing Hardcoded Values with Constants:

PROMPT:
Refactor the Python code by replacing hardcoded numbers
with descriptive constants to improve readability.

print("Simple Interest:", (1000 * 2 * 5) / 100)
print("Simple Interest:", (2000 * 2 * 5) / 100)

CODE:

```python
RATE = 2
TIME = 5

def calculate_simple_interest(principal):
    interest = (principal * RATE * TIME) / 100
    print("Simple Interest:", interest)


calculate_simple_interest(1000)
calculate_simple_interest(2000)
```

OUPUT:

```
sushruthchada@sushruths-MacBook-Air-2 coding % /usr/local/bin/python3 "/U
Simple Interest: 100.0
Simple Interest: 200.0
sushruthchada@sushruths-MacBook-Air-2 coding %
```

Task 5 – Improving Variable Naming:

PROMPT:
Improve the readability of the following Python code
by renaming variables and adding meaningful comments.

x = 50
y = 30
z = x + y
print(z)

CODE:

```python
# first number
num1 = 50

# second number
num2 = 30

# calculate sum
total_sum = num1 + num2

# print result
print(total_sum)
```

OUTPUT:

```
sushruthchada@sushruths-MacBook-Air-2 coding % /usr/local/bin/python3
80
sushruthchada@sushruths-MacBook-Air-2 coding %
```